

CA2302: Ensure BinaryFormatter.Binder is set before calling BinaryFormatter.Deserialize (code analysis)

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Content

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CA2302: Ensure BinaryFormatter.Binder is set before calling BinaryFormatter.Deserialize

Summarize this article for me

| Property | Value | | **Rule ID** | CA2302 | | **Title** | Ensure BinaryFormatter.Binder is set before calling BinaryFormatter.Deserialize | | **Category** | [Security](#) | | **Fix is breaking or non-breaking** | Non-breaking | | **Enabled by default in .NET 10** | No | | **Applicable languages** | C# and Visual Basic | |

Cause

A [System.Runtime.Serialization.Formatters.Binary.BinaryFormatter](#) deserialization method was called or referenced and the [Binder](#) property may be null.

This rule is similar to [CA2301](#), but analysis can't determine if the [Binder](#) is definitely null. By default, this rule analyzes the entire codebase, but this is [configurable]().

Warning

Restricting types with a `SerializationBinder` can't prevent all attacks. For more information, see the [BinaryFormatter security guide](#).

Rule description

Insecure deserializers are vulnerable when deserializing untrusted data. An attacker could modify the serialized data to include unexpected types to inject objects with malicious side effects. An attack against an insecure deserializer could, for example, execute commands on the underlying operating system, communicate over the network, or delete files.

This rule finds `System.Runtime.Serialization.Formatters.Binary.BinaryFormatter` deserialization method calls or references when the `Binder` might be null. If you want to disallow any deserialization with `BinaryFormatter` regardless of the `Binder` property, disable this rule and [CA2301](#), and enable rule [CA2300](#).

How to fix violations

- Use a secure serializer instead, and **don't allow an attacker to specify an arbitrary type to deserialize**. For more information see the [Preferred alternatives](#).
- Make the serialized data tamper-proof. After serialization, cryptographically sign the serialized data. Before deserialization, validate the cryptographic signature. Protect the cryptographic key from being disclosed and design for key rotations.
- This option makes code vulnerable to denial of service attacks and possible remote code execution attacks in the future. For more information, see the [BinaryFormatter security guide](#). Restrict deserialized types. Implement a custom `System.Runtime.Serialization.on.SerializationBinder`. Before deserializing, set the `Binder` property to an instance of your custom `SerializationBinder` in all code paths. In the overridden `BindToType` method, if the type is unexpected, throw an exception to stop deserialization.

When to suppress warnings

`BinaryFormatter` is insecure and can't be made secure.

Configure code to analyze

Use the following options to configure which parts of your codebase to run this rule on.

- `[Exclude specific symbols]()`
- `[Exclude specific types and their derived types]()`

You can configure these options for just this rule, for all rules they apply to, or for all rules in this category ([Security](#)) that they apply to. For more information, see [Code quality rule configuration options](#).

Exclude specific symbols

You can exclude specific symbols, such as types and methods, from analysis by setting the [excluded_symbol_names](#) option. For example, to specify that the rule should not run on any code within types named `MyType`, add the following key-value pair to an `.editorconfig` file in your project:

```
dotnet_code_quality.CAXXXX.excluded_symbol_names = MyType
```

Note

Replace the `XXXX` part of `CAXXXX` with the ID of the applicable rule.

Allowed symbol name formats in the option value (separated by `|`):

- Symbol name only (includes all symbols with the name, regardless of the containing type or namespace).
- Fully qualified names in the symbol's [documentation ID format](#). Each symbol name requires a symbol-kind prefix, such as `M:` for methods, `T:` for types, and `N:` for namespaces.
- `.ctor` for constructors and `.cctor` for static constructors.

Examples:

Option Value	Summary
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = MyType</code>	Matches all symbols named <code>MyType</code> .

<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = MyType1 MyType2</code>	Matches all symbols named either <code>MyType1</code> or <code>MyType2</code> .
---	---

<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = M:NS.MyType.MyMethod(ParamType)</code>	Matches specific method <code>MyMethod</code> with the specified fully qualified signature.
---	---

<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = M:NS1.MyType1.MyMethod1(ParamType) M:NS2.MyType2.MyMethod2(ParamType)</code>	Matches specific methods <code>MyMethod1</code> and <code>MyMethod2</code> with the respective fully qualified signatures.
---	--

Exclude specific types and their derived types

You can exclude specific types and their derived types from analysis by setting the [exclude_type_names_with_derived_types](#) option. For example, to specify that the rule should not run on any methods within types named `MyType` and their derived types, add the following key-value pair to an `.editorconfig` file in your project:

```
dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = MyType
```

Note

Replace the `XXXX` part of `CAXXXX` with the ID of the applicable rule.

Allowed symbol name formats in the option value (separated by `|`):

- Type name only (includes all types with the name, regardless of the containing type or namespace).

- Fully qualified names in the symbol's [documentation ID format](#), with an optional `T:` prefix.

Examples:

Option value	Summary
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = MyType</code>	Matches all types named <code>MyType</code> and all of their derived types.
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = MyType1 MyType2</code>	Matches all types named either <code>MyType1</code> or <code>MyType2</code> and all of their derived types.
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = M:NS.MyType</code>	Matches specific type <code>MyType</code> with given fully qualified name and all of its derived types.
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = M:NS1.MyType1 M:NS2.MyType2</code>	Matches specific types <code>MyType1</code> and <code>MyType2</code> with the respective fully qualified names, and all of their derived types.

Pseudo-code examples

Violation 1

```

using System;
using System.IO;
using System.Runtime.Serialization;
using System.Runtime.Serialization.Formatters.Binary;

public class BookRecordSerializationBinder : SerializationBinder
{
    public override Type BindToType(string assemblyName, string typeName)
    {
        // One way to discover expected types is through testing deserialization
        // of **valid** data and logging the types used.

        ////Console.WriteLine($"BindToType('{assemblyName}', '{typeName}')");

        if (typeName == "BookRecord")
        {
            return typeof(BookRecord);
        }
        else if (typeName == "AisleLocation")
        {
            return typeof(AisleLocation);
        }
        else
        {
            throw new ArgumentException("Unexpected type", nameof(typeName));
        }
    }
}

[Serializable]
public class BookRecord
{
    public string Title { get; set; }
    public AisleLocation Location { get; set; }
}

[Serializable]
public class AisleLocation
{
    public char Aisle { get; set; }
    public byte Shelf { get; set; }
}

public class Binders

```

```
{  
    public static SerializationBinder BookRecord =  
        new BookRecordSerializationBinder();  
}  
  
public class ExampleClass  
{  
    public BookRecord DeserializeBookRecord(byte[] bytes)  
    {  
        BinaryFormatter formatter = new BinaryFormatter();  
        formatter.Binder = Binders.BookRecord;  
        using (MemoryStream ms = new MemoryStream(bytes))  
        {  
            return (BookRecord)formatter.Deserialize(ms);    // CA2302 violation  
        }  
    }  
}
```

```

Imports System
Imports System.IO
Imports System.Runtime.Serialization
Imports System.Runtime.Serialization.Formatters.Binary

Public Class BookRecordSerializationBinder
    Inherits SerializationBinder

    Public Overrides Function BindToType(assemblyName As String, typeName As String) As Type
        ' One way to discover expected types is through testing deserialization
        ' of **valid** data and logging the types used.

        Console.WriteLine($"BindToType('{assemblyName}', '{typeName}')" )

        If typeName = "BinaryFormatterVB.BookRecord" Then
            Return GetType(BookRecord)
        Else If typeName = "BinaryFormatterVB.AisleLocation" Then
            Return GetType(AisleLocation)
        Else
            Throw New ArgumentException("Unexpected type", NameOf(typeName))
        End If
    End Function
End Class

<Serializable(>
Public Class BookRecord
    Public Property Title As String
    Public Property Location As AisleLocation
End Class

<Serializable(>
Public Class AisleLocation
    Public Property Aisle As Char
    Public Property Shelf As Byte
End Class

Public Class Binders
    Public Shared Property BookRecord As SerializationBinder = New
BookRecordSerializationBinder()
End Class

Public Class ExampleClass
    Public Function DeserializeBookRecord(bytes As Byte()) As BookRecord

```

```
Dim formatter As BinaryFormatter = New BinaryFormatter()  
formatter.Binder = Binders.BookRecord  
Using ms As MemoryStream = New MemoryStream(bytes)  
    Return CType(formatter.Deserialize(ms), BookRecord)    ' CA2302 violation  
End Using  
End Function  
End Class
```

Solution 1

```

using System;
using System.IO;
using System.Runtime.Serialization;
using System.Runtime.Serialization.Formatters.Binary;

public class BookRecordSerializationBinder : SerializationBinder
{
    public override Type BindToType(string assemblyName, string typeName)
    {
        // One way to discover expected types is through testing deserialization
        // of **valid** data and logging the types used.

        ////Console.WriteLine($"BindToType('{assemblyName}', '{typeName}')");

        if (typeName == "BookRecord")
        {
            return typeof(BookRecord);
        }
        else if (typeName == "AisleLocation")
        {
            return typeof(AisleLocation);
        }
        else
        {
            throw new ArgumentException("Unexpected type", nameof(typeName));
        }
    }
}

[Serializable]
public class BookRecord
{
    public string Title { get; set; }
    public AisleLocation Location { get; set; }
}

[Serializable]
public class AisleLocation
{
    public char Aisle { get; set; }
    public byte Shelf { get; set; }
}

public class Binders

```

```
{  
    public static SerializationBinder BookRecord =  
        new BookRecordSerializationBinder();  
}  
  
public class ExampleClass  
{  
    public BookRecord DeserializeBookRecord(byte[] bytes)  
    {  
        BinaryFormatter formatter = new BinaryFormatter();  
  
        // Ensure that Binder is always non-null before deserializing  
        formatter.Binder = Binders.BookRecord ?? throw new Exception("Expected non-null  
binder");  
  
        using (MemoryStream ms = new MemoryStream(bytes))  
        {  
            return (BookRecord)formatter.Deserialize(ms);  
        }  
    }  
}
```

```

Imports System
Imports System.IO
Imports System.Runtime.Serialization
Imports System.Runtime.Serialization.Formatters.Binary

Public Class BookRecordSerializationBinder
    Inherits SerializationBinder

    Public Overrides Function BindToType(assemblyName As String, typeName As String) As Type
        ' One way to discover expected types is through testing deserialization
        ' of **valid** data and logging the types used.

        Console.WriteLine($"BindToType('{assemblyName}', '{typeName}')" )

        If typeName = "BinaryFormatterVB.BookRecord" Then
            Return GetType(BookRecord)
        Else If typeName = "BinaryFormatterVB.AisleLocation" Then
            Return GetType(AisleLocation)
        Else
            Throw New ArgumentException("Unexpected type", NameOf(typeName))
        End If
    End Function
End Class

<Serializable(>
Public Class BookRecord
    Public Property Title As String
    Public Property Location As AisleLocation
End Class

<Serializable(>
Public Class AisleLocation
    Public Property Aisle As Char
    Public Property Shelf As Byte
End Class

Public Class Binders
    Public Shared Property BookRecord As SerializationBinder = New
BookRecordSerializationBinder()
End Class

Public Class ExampleClass
    Public Function DeserializeBookRecord(bytes As Byte()) As BookRecord

```

```

Dim formatter As BinaryFormatter = New BinaryFormatter()

' Ensure that Binder is always non-null before deserializing
formatter.Binder = If(Binders.BookRecord, New Exception("Expected non-null"))

Using ms As MemoryStream = New MemoryStream(bytes)
    Return CType(formatter.Deserialize(ms), BookRecord)
End Using
End Function
End Class

```

Violation 2

```

using System;
using System.IO;
using System.Runtime.Serialization.Formatters.Binary;

[Serializable]
public class BookRecord
{
    public string Title { get; set; }
    public AisleLocation Location { get; set; }
}

[Serializable]
public class AisleLocation
{
    public char Aisle { get; set; }
    public byte Shelf { get; set; }
}

public class ExampleClass
{
    public BinaryFormatter Formatter { get; set; }

    public BookRecord DeserializeBookRecord(byte[] bytes)
    {
        using (MemoryStream ms = new MemoryStream(bytes))
        {
            return (BookRecord) this.Formatter.Deserialize(ms); // CA2302 violation
        }
    }
}

```

```

Imports System
Imports System.IO
Imports System.Runtime.Serialization.Formatters.Binary

<Serializable(>>
Public Class BookRecord
    Public Property Title As String
    Public Property Location As AisleLocation
End Class

<Serializable(>>
Public Class AisleLocation
    Public Property Aisle As Char
    Public Property Shelf As Byte
End Class

Public Class ExampleClass
    Public Property Formatter As BinaryFormatter

    Public Function DeserializeBookRecord(bytes As Byte()) As BookRecord
        Using ms As MemoryStream = New MemoryStream(bytes)
            Return CType(Me.Formatter.Deserialize(ms), BookRecord) ' CA2302 violation
        End Using
    End Function
End Class

```

Solution 2

```

using System;
using System.IO;
using System.Runtime.Serialization;
using System.Runtime.Serialization.Formatters.Binary;

public class BookRecordSerializationBinder : SerializationBinder
{
    public override Type BindToType(string assemblyName, string typeName)
    {
        // One way to discover expected types is through testing deserialization
        // of **valid** data and logging the types used.

        ////Console.WriteLine($"BindToType('{assemblyName}', '{typeName}')");

        if (typeName == "BookRecord")
        {
            return typeof(BookRecord);
        }
        else if (typeName == "AisleLocation")
        {
            return typeof(AisleLocation);
        }
        else
        {
            throw new ArgumentException("Unexpected type", nameof(typeName));
        }
    }
}

[Serializable]
public class BookRecord
{
    public string Title { get; set; }
    public AisleLocation Location { get; set; }
}

[Serializable]
public class AisleLocation
{
    public char Aisle { get; set; }
    public byte Shelf { get; set; }
}

public class ExampleClass

```

```
{  
    public BookRecord DeserializeBookRecord(byte[] bytes)  
    {  
        BinaryFormatter formatter = new BinaryFormatter();  
        formatter.Binder = new BookRecordSerializationBinder();  
        using (MemoryStream ms = new MemoryStream(bytes))  
        {  
            return (BookRecord) formatter.Deserialize(ms);  
        }  
    }  
}
```

```

Imports System
Imports System.IO
Imports System.Runtime.Serialization
Imports System.Runtime.Serialization.Formatters.Binary

Public Class BookRecordSerializationBinder
    Inherits SerializationBinder

    Public Overrides Function BindToType(assemblyName As String, typeName As String) As Type
        ' One way to discover expected types is through testing deserialization
        ' of **valid** data and logging the types used.

        Console.WriteLine($"BindToType('{assemblyName}', '{typeName}')" )

        If typeName = "BinaryFormatterVB.BookRecord" Then
            Return GetType(BookRecord)
        Else If typeName = "BinaryFormatterVB.AisleLocation" Then
            Return GetType(AisleLocation)
        Else
            Throw New ArgumentException("Unexpected type", NameOf(typeName))
        End If
    End Function
End Class

<Serializable(>>
Public Class BookRecord
    Public Property Title As String
    Public Property Location As AisleLocation
End Class

<Serializable(>>
Public Class AisleLocation
    Public Property Aisle As Char
    Public Property Shelf As Byte
End Class

Public Class ExampleClass
    Public Function DeserializeBookRecord(bytes As Byte()) As BookRecord
        Dim formatter As BinaryFormatter = New BinaryFormatter()
        formatter.Binder = New BookRecordSerializationBinder()
        Using ms As MemoryStream = New MemoryStream(bytes)
            Return CType(formatter.Deserialize(ms), BookRecord)
        End Using
    End Function
End Class

```

End **Function**

End **Class**

- CA2300: Do not use insecure deserializer BinaryFormatter
- CA2301: Do not call BinaryFormatter.Deserialize without first setting BinaryFormatter.Binder

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